

ALEX NGECHU MUIRURI

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WORK EXPERIENCE

Gearbox Kenya

Engineering Intern

JAN 2023 - MAR 2024

- Collaborated with senior engineers to analyze 50+ data points and build 3 computer models, resulting in 15% optimization of product designs for smart farms.
- Conducted a comprehensive research project on renewable energy sources, providing actionable recommendations for implementation, which led to a 20% reduction in carbon emissions.

Kenya Airport Authority

Student Intern

FEB 2024 - APR 2024

- Assisted in the maintenance and troubleshooting of airport equipment and infrastructure systems, including passenger boarding bridges, resulting in a 25% reduction in downtime.
- Supported senior engineers in performing regular maintenance on critical infrastructure, gaining hands-on experience with PLC programming, compliance procedures, and safety protocols within the aviation industry.
- Contributed to maintaining system reliability and safety for airport operations, ensuring seamless travel experiences for over 10,000 passengers daily.

Africa Youth Leadership Forum

student leader

APR 2024 - DEC 2024

- Spearheaded a community development project that engaged over 200 students in mentorship and mental health awareness among high school students, resulting in a 30% reduction in reported mental health issues among participants.

EDUCATION

Jomo Kenyatta University of Agriculture and Technology

Mechatronic engineering

2019 - 2024

AWS

Machine learning for business

DEC 2024 - Feb 2025

PROJECTS

PLC BASED PLASTIC EXTRUSION MACHINE

- Designed and implemented a PLC-based control system for a plastic extrusion machine.
- Developed temperature control logic (270-280C) using both ON/OFF and analog control methods.
- Focused on efficient temperature distribution within the barrel to enhance extrusion quality.
- Achieved hydraulic control for the actuation of clamping and unclamping.

SKILLS

- PLC programming
- CAD/CAM
- Python
- Java
- Autodesk
- Robotics

CERTIFICATE

AWS Machine learning ready organization

- Certified in AWS Machine Learning, demonstrating expertise in deploying machine learning models on scalable and secure cloud infrastructure.